

BROCKET AGDM, AB, Canada

WMO: 713410

Lat: 49.6107N

Lon: 113.7591W

Elev: 3519

StdP: 12.92

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.4	-9.4	-22.0	1.9	-15.2	-15.4	2.7	-8.9	43.6	30.9	40.3	33.1	7.1	310	0.772

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.8	87.7	60.2	84.1	59.3	80.7	58.7	64.3	79.0	62.3	78.0	60.5	75.9	11.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	84.8	71.2	56.5	77.6	68.0	54.5	72.0	65.2	31.3	78.5	29.7	78.1	28.4	76.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.0	35.0	30.2	DB	-22.8	93.5	9.4	3.8	-29.6	96.3	-35.1	98.5	-40.3	100.6	-47.1	103.4	(4)
(5)			WB	-23.2	69.3	9.3	2.8	-29.9	71.3	-35.3	73.0	-40.5	74.6	-47.2	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	25.1	23.7	31.1	41.0	50.0	57.4	64.3	63.2	54.7	43.3	32.7	24.4	(6)	
	DBStd	18.08	14.85	14.58	13.82	9.10	8.09	5.63	5.49	6.11	8.60	9.96	12.49	13.81	(7)	
	HDD50	4104	772	737	587	289	101	7	0	2	48	246	523	792	(8)	
	HDD65	8266	1235	1157	1050	721	466	236	79	105	320	671	969	1257	(9)	
	CDD50	1437	1	0	2	18	102	228	442	412	189	40	3	0	(10)	
	CDD65	126	0	0	0	0	2	8	56	50	10	0	0	0	(11)	
	CDH74	2775	0	0	0	6	91	216	1109	1032	310	11	0	0	(12)	
	CDH80	867	0	0	0	0	16	44	379	346	81	1	0	0	(13)	
(14) Wind	WSAvg	12.9	16.8	13.4	13.6	12.7	11.7	11.1	9.5	9.1	11.3	13.7	15.8	15.9	(14)	
(15) Precipitation	PrecAvg	21.1	0.8	0.8	1.1	1.7	3.0	4.5	2.0	1.8	1.8	1.0	1.0	0.9	(15)	
	PrecMax	34.2	1.5	1.9	2.6	3.5	6.1	14.8	7.4	4.8	5.9	2.0	2.3	2.3	(16)	
	PrecMin	12.0	0.2	0.0	0.3	0.5	0.2	1.3	0.0	0.0	0.2	0.1	0.2	0.2	(17)	
	PrecStd	5.9	0.3	0.5	0.6	0.7	1.7	3.0	1.8	1.3	1.3	0.6	0.6	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	54.3	61.5	73.3	82.3	85.7	92.7	91.5	86.9	73.7	62.3	50.8	(19)	
		MCWB	42.7	40.4	44.6	49.3	55.4	62.4	61.8	59.5	57.3	53.1	46.4	41.1	(20)	
	2%	DB	49.2	49.1	56.0	66.7	76.0	79.4	88.1	87.5	82.3	67.4	55.4	47.0	(21)	
		MCWB	39.5	38.3	42.5	47.5	53.3	59.7	61.5	59.2	56.1	50.3	43.9	38.5	(22)	
	5%	DB	45.6	45.2	52.4	61.5	71.0	75.7	84.3	83.9	77.3	62.3	51.3	43.8	(23)	
		MCWB	37.2	36.0	40.8	44.9	51.5	57.9	60.9	58.4	55.0	48.4	41.8	36.6	(24)	
	10%	DB	42.2	41.9	48.4	56.2	66.2	71.6	80.4	80.1	71.6	57.7	47.6	41.2	(25)	
		MCWB	35.2	34.1	38.7	42.4	49.5	55.8	60.5	57.8	52.9	45.7	39.5	34.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	42.1	46.5	51.4	57.7	65.7	68.2	64.2	60.2	54.6	47.8	42.7	(27)	
		MCDB	52.9	51.0	58.7	70.2	76.4	80.9	81.7	80.5	78.7	68.8	59.1	49.7	(28)	
	2%	WB	40.1	39.1	43.7	48.3	54.5	62.1	65.3	62.3	57.6	51.3	44.9	39.5	(29)	
		MCDB	48.1	48.2	54.1	64.3	71.7	75.9	79.9	79.7	77.4	65.2	54.3	45.4	(30)	
	5%	WB	37.6	36.5	41.2	45.8	52.4	59.3	63.4	60.7	55.7	48.8	42.1	37.2	(31)	
		MCDB	44.1	44.4	51.0	59.8	68.6	72.2	78.4	77.3	73.9	60.9	50.4	43.2	(32)	
	10%	WB	35.7	34.2	39.0	43.2	50.5	57.1	61.6	59.1	53.7	46.2	39.7	34.9	(33)	
		MCDB	42.0	41.6	48.0	54.9	64.6	69.3	76.8	76.1	70.6	56.7	47.3	41.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	19.2	19.5	21.2	22.2	22.3	27.8	28.1	24.0	20.0	16.3	16.1	(35)	
		MCDBR	17.1	21.2	23.9	30.4	31.3	30.7	35.2	35.8	34.7	25.6	18.6	15.3	(36)	
		MCWBR	11.1	13.2	13.4	15.1	13.7	14.8	14.5	13.3	14.0	12.8	11.2	10.4	(37)	
	5% WB	MCDBR	16.5	21.3	22.5	28.7	28.0	28.0	30.0	30.6	31.8	25.2	18.3	16.1	(38)	
		MCWBR	11.9	14.0	13.3	14.8	12.8	14.7	14.7	13.0	13.6	13.1	11.5	11.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.250	0.259	0.285	0.333	0.340	0.334	0.336	0.351	0.316	0.285	0.262	0.234	(40)	
	taud		2.405	2.420	2.438	2.334	2.388	2.446	2.449	2.394	2.470	2.532	2.488	2.428	(41)	
	Ebn at Noon		262	288	296	290	291	293	291	280	282	274	256	255	(42)	
	Edh at Noon		19	25	29	36	36	34	34	34	28	22	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		382	697	1078	1506	1794	1982	2213	1774	1253	773	422	302	(44)	
	RadStd		33	54	92	106	151	149	130	114	111	79	35	29	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page